

## Lecture 12

### Tensors and the Tensor Product

Lecture 11 introduced the indices; this lecture builds the objects they live on. The *tensor product*  $V \otimes W$  is the vector space whose elements combine a vector from  $V$  with one from  $W$  multiplicatively, and tensors of higher type unify vectors, covectors, operators, and bilinear forms into one framework. Its matrix incarnation is the *Kronecker product*, and its physical meaning is decisive: the state space of a composite quantum system is the tensor product of its parts, and *entanglement* is precisely a state that fails to factor. This closes Part II of the course.

#### Learning objectives

By the end of this lecture you will be able to:

- ▶ Construct  $V \otimes W$  via its basis and universal property; spot non-factoring tensors.
- ▶ Compute Kronecker products and use  $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$ .
- ▶ Read a type- $(p, q)$  tensor as a multilinear map and contract indices.
- ▶ Distinguish product from entangled states in composite quantum systems.

#### 1 The tensor product

A bilinear map  $V \times W \rightarrow U$  is awkward—it is not linear. The tensor product repackages it as an ordinary linear map out of a single space.

**Definition 12.1 (Tensor product).** The *tensor product*  $V \otimes W$  is the vector space with basis  $\{e_i \otimes f_j\}$ , where  $e_i$  is a basis of  $V$  and  $f_j$  a basis of  $W$ . Thus

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

For  $v \in V, w \in W$  the *elementary* (or decomposable) tensor  $v \otimes w$  is bilinear:  $(v + v') \otimes w = v \otimes w + v' \otimes w$  and  $(av) \otimes w = a(v \otimes w) = v \otimes (aw)$ .

A general element of  $V \otimes W$  is a sum  $\sum_{ij} \tau^{ij} e_i \otimes f_j$  of elementary tensors; crucially, most tensors are *not* themselves elementary—this is the algebraic seed of entanglement.

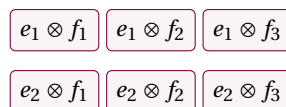


Figure 1: A basis of  $V \otimes W$  for  $\dim V = 2, \dim W = 3$ : every pair  $(e_i, f_j)$  contributes one basis tensor, giving  $\dim(V \otimes W) = 2 \cdot 3 = 6$ .

**Example 12.2 (A tensor that does not factor).** In  $\mathbb{C}^2 \otimes \mathbb{C}^2$  the tensor  $e_1 \otimes e_1 + e_2 \otimes e_2$  is not elementary: if it equalled  $v \otimes w$  with  $v = (a, b), w = (c, d)$ , its components would be the rank-one array  $\begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$ , whereas the given tensor has components  $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ , of rank two. A tensor is decomposable exactly when its component matrix has rank one. ◀

**Example 12.3 (Functions of two variables).** The tensor product of polynomial spaces realizes products of functions:  $\mathcal{P}_m(x) \otimes \mathcal{P}_n(y)$  is spanned by  $x^i \otimes y^j$ , identified with the monomials

$x^i y^j$ . A general element—a polynomial in  $x$  and  $y$ —is a sum of products  $f(x)g(y)$ , and is a single product exactly when it factors. ◀

**Remark 12.4 (The universal property).** The defining feature of  $V \otimes W$  is that every bilinear map  $B: V \times W \rightarrow U$  factors uniquely as  $B(v, w) = \tilde{B}(v \otimes w)$  for a unique linear  $\tilde{B}: V \otimes W \rightarrow U$ . The tensor product is the universal recipient of bilinear maps—the precise sense in which it “linearizes multiplication.”

**Remark 12.5 (Many factors).** The tensor product is associative,  $U \otimes (V \otimes W) = (U \otimes V) \otimes W$ , so multi-factor products  $V_1 \otimes \cdots \otimes V_p$  need no parentheses; their basis is the products  $e_{i_1} \otimes \cdots \otimes e_{i_p}$  and their dimension is the product of the dimensions. This is the home of  $(p, q)$  tensors and of many-body quantum states.

## 2 The Kronecker product

In coordinates the tensor product is the *Kronecker product*: stacking the basis  $e_i \otimes f_j$  lexicographically, the column of  $v \otimes w$  is the Kronecker product of the columns of  $v$  and  $w$ , and an operator  $A \otimes B$  has the block form

$$A \otimes B = \begin{pmatrix} A_{11}B & \cdots & A_{1n}B \\ \vdots & & \vdots \\ A_{n1}B & \cdots & A_{nn}B \end{pmatrix},$$

each entry  $A_{ij}$  of  $A$  replaced by the block  $A_{ij}B$ .

**Proposition 12.6 (Properties of  $A \otimes B$ ).** The Kronecker product satisfies

$$(A \otimes B)(v \otimes w) = (Av) \otimes (Bw), \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD),$$

together with  $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ ,  $\text{tr}(A \otimes B) = (\text{tr } A)(\text{tr } B)$ , and if  $Av = \lambda v$ ,  $Bw = \mu w$  then  $(A \otimes B)(v \otimes w) = \lambda\mu(v \otimes w)$ .

*Proof.* By linearity it suffices to check the identities on elementary tensors. From the block form,  $(A \otimes B)(e_i \otimes f_j) = \sum_{a,b} A_{ai} B_{bj} e_a \otimes f_b = (Ae_i) \otimes (Bf_j)$ , so  $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$ . The composition and eigenvalue identities follow at once:  $(A \otimes B)(C \otimes D)(v \otimes w) = (ACv) \otimes (BDw) = ((AC) \otimes (BD))(v \otimes w)$ , and  $(A \otimes B)(v \otimes w) = (\lambda v) \otimes (\mu w) = \lambda\mu(v \otimes w)$  when  $Av = \lambda v$ ,  $Bw = \mu w$ . For the adjoint, using  $\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \langle w, w' \rangle$  and  $\langle Av, x \rangle = \langle v, A^\dagger x \rangle$ ,

$$\begin{aligned} \langle (A \otimes B)(v \otimes w), x \otimes y \rangle &= \langle Av, x \rangle \langle Bw, y \rangle = \langle v, A^\dagger x \rangle \langle w, B^\dagger y \rangle \\ &= \langle v \otimes w, (A^\dagger \otimes B^\dagger)(x \otimes y) \rangle, \end{aligned}$$

so  $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ . Finally, summing the diagonal of the block form,  $\text{tr}(A \otimes B) = \sum_{a,b} A_{aa} B_{bb} = (\text{tr } A)(\text{tr } B)$ . ■

So eigenvalues multiply: the spectrum of  $A \otimes B$  is  $\{\lambda_i \mu_j\}$ , and if  $A, B$  are each diagonalizable (or normal) then so is  $A \otimes B$ .

**Example 12.7 (A Kronecker product of Pauli matrices).** With  $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

the operator that flips both qubits. Its eigenvalues are the products  $(\pm 1)(\pm 1)$ , namely  $+1$  (twice, on  $|+\rangle \otimes |+\rangle$  and  $|-\rangle \otimes |-\rangle$ ) and  $-1$  (twice, on  $|+\rangle \otimes |-\rangle$  and  $|-\rangle \otimes |+\rangle$ )—consistent with  $\text{tr}(\sigma_x \otimes \sigma_x) = (\text{tr} \sigma_x)^2 = 0$ .  $\triangleleft$

**Example 12.8 (Kronecker product of vectors).** For  $v = (1, 2)$  and  $w = (3, 4)$ ,

$$v \otimes w = (1 \cdot w, 2 \cdot w) = (3, 4, 6, 8),$$

the four products  $v_i w_j$  in lexicographic order. The two-qubit state  $|0\rangle \otimes |1\rangle$  is likewise the column  $(0, 1, 0, 0) = |01\rangle$ .  $\triangleleft$

**Example 12.9 (Acting factorwise).** With  $A = \sigma_x$ ,  $B = \sigma_z$  and  $|00\rangle = |0\rangle \otimes |0\rangle$ ,  $(\sigma_x \otimes \sigma_z)|00\rangle = (\sigma_x|0\rangle) \otimes (\sigma_z|0\rangle) = |1\rangle \otimes |0\rangle = |10\rangle$ : each factor acts on its own slot, the content of  $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$ .  $\triangleleft$

**Example 12.10 (A single-qubit gate on two qubits).** To apply a Hadamard  $H$  to the first qubit only, use  $H \otimes I$ : it acts as  $H$  on the first slot and leaves the second alone,  $(H \otimes I)|0x\rangle = |+\rangle \otimes |x\rangle$ . Local gates are tensor products with identities, and they cannot create entanglement.  $\triangleleft$

**Example 12.11 (A two-qubit gate matrix).** The Hadamard on both qubits is

$$H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

each scalar entry  $\frac{1}{\sqrt{2}} H_{ij}$  scaling the block  $\frac{1}{\sqrt{2}} H$ . Applied to  $|00\rangle$  it produces the uniform superposition  $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ .  $\triangleleft$

|           |           |
|-----------|-----------|
| $A_{11}B$ | $A_{12}B$ |
| $A_{21}B$ | $A_{22}B$ |

Figure 2: The Kronecker product  $A \otimes B$  for a  $2 \times 2$  matrix  $A$ : each scalar entry  $A_{ij}$  becomes the block  $A_{ij}B$ , building an  $(nm) \times (nm)$  matrix from  $n \times n$  and  $m \times m$  pieces.

**Example 12.12 (The mixed-product rule).** The rule  $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$  lets gates compose factorwise. For instance  $(\sigma_x \otimes I)(\sigma_z \otimes I) = (\sigma_x \sigma_z) \otimes I = (-i\sigma_y) \otimes I$ , so applying  $\sigma_z$  then  $\sigma_x$  to the first qubit is the single gate  $-i\sigma_y \otimes I$ .  $\triangleleft$

**Remark 12.13 (The swap operator).** The swap  $|ij\rangle \mapsto |ji\rangle$  exchanges the two factors of  $\mathcal{H} \otimes \mathcal{H}$ . It acts as  $+1$  on the symmetric tensors  $S^2\mathcal{H}$  and  $-1$  on the antisymmetric tensors  $\Lambda^2\mathcal{H}$  (both defined in §4 below), so the  $S^2 \oplus \Lambda^2$  split is exactly its eigenspace decomposition.

**Example 12.14 (Single-qubit operators on two qubits).** On two qubits  $\sigma_z \otimes I$  and  $I \otimes \sigma_z$  are the diagonal matrices

$$\sigma_z \otimes I = \text{diag}(1, 1, -1, -1), \quad I \otimes \sigma_z = \text{diag}(1, -1, 1, -1),$$

measuring the first and second spin respectively. They commute, and their product is  $\sigma_z \otimes \sigma_z = \text{diag}(1, -1, -1, 1)$ , the parity.  $\triangleleft$

**Remark 12.15 (Trace, determinant, and size).** From  $\text{tr}(A \otimes B) = \text{tr} A \text{tr} B$  and the multiplied eigenvalues,  $\det(A \otimes B) = (\det A)^m (\det B)^n$  for  $A$  of size  $n$  and  $B$  of size  $m$ . The dimensions check out:  $A \otimes B$  is  $nm \times nm$ , matching  $\dim(V \otimes W) = nm$ .

**Remark 12.16 (Inner product on a tensor product).** If  $V, W$  are inner product spaces,  $V \otimes W$  inherits the inner product  $\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \langle w, w' \rangle$  and an orthonormal basis  $\{e_i \otimes f_j\}$ . Composite quantum systems are therefore again inner product (Hilbert) spaces, so everything in Lectures 7–10 applies to them.

### 3 Tensors as multilinear maps

A *tensor of type  $(p, q)$*  is a multilinear map taking  $p$  covectors and  $q$  vectors to a scalar; equivalently it is an element of  $V^{\otimes p} \otimes (V^*)^{\otimes q}$ , with components  $\tau^{i_1 \dots i_p}_{j_1 \dots j_q}$  carrying  $p$  upper and  $q$  lower indices. The familiar objects are low-rank tensors: a vector is  $(1, 0)$ , a covector  $(0, 1)$ , a bilinear form (such as the metric  $g_{ij}$ ) is  $(0, 2)$ , and a linear operator is  $(1, 1)$ .

**Proposition 12.17 (Operators are tensors).** There is a canonical isomorphism

$$\mathcal{L}(V, W) \cong W \otimes V^*, \quad A^a{}_i \longmapsto A^a{}_i w_a \otimes e^i,$$

sending an operator to the tensor built from its matrix. The identity operator corresponds to  $\delta^a{}_i w_a \otimes e^i$ —the Kronecker delta is the identity tensor.

*Proof.* Define  $\Psi: W \otimes V^* \rightarrow \mathcal{L}(V, W)$  on elementary tensors by  $\Psi(w \otimes \varphi)(v) = \varphi(v) w$ , extended linearly; this uses only the pairing of  $V^*$  with  $V$ , so it needs no basis. On the basis  $\{w_a \otimes e^i\}$ ,  $\Psi(w_a \otimes e^i)(e_j) = e^i(e_j) w_a = \delta^i_j w_a$ , the elementary operator  $E^a{}_i$  sending  $e_i \mapsto w_a$  and  $e_j \mapsto 0$  ( $j \neq i$ ). As  $\{E^a{}_i\}$  is a basis of  $\mathcal{L}(V, W)$ ,  $\Psi$  carries a basis to a basis and is an isomorphism, with  $\dim = (\dim W)(\dim V)$ . Its inverse sends  $A = A^a{}_i$  to  $A^a{}_i w_a \otimes e^i$ , the correspondence of the statement; that tensor returns  $v \mapsto A^a{}_i e^i(v) w_a = A^a{}_i v^i w_a = Av$ . ■

**Example 12.18 (Outer products are decomposable operators).** The rank-one operator  $|a\rangle\langle b|$  corresponds to the elementary tensor  $|a\rangle \otimes \langle b| \in V \otimes V^*$ . A general operator is a sum of such outer products (its tensor is a sum of decomposables), and it is itself decomposable—rank one—exactly when it is a single outer product  $|a\rangle\langle b|$ . ◀

**Example 12.19 (The metric as a  $(0, 2)$  tensor).** The inner product is the  $(0, 2)$  tensor  $g = g_{ij} e^i \otimes e^j$  with  $g_{ij} = \langle e_i, e_j \rangle$ ; it eats two vectors and returns  $g(u, v) = g_{ij} u^i v^j$ . Lowering an index,  $v_j = g_{ij} v^i$ , is the contraction of  $g$  with a vector—the index-lowering of Lecture 11. ◀

**Example 12.20 (An operator as components).** The operator  $T = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  has  $(1, 1)$  components  $T^1{}_1 = 2, T^1{}_2 = 1, T^2{}_1 = 0, T^2{}_2 = 3$ , and as a tensor  $T = 2 e_1 \otimes e^1 + e_1 \otimes e^2 + 3 e_2 \otimes e^2$ . Its trace is the contraction  $T^i{}_i = 2 + 3 = 5$ . ◀

**Example 12.21 (Trace via the identity tensor).** Contracting the identity tensor  $\delta^i{}_j$  with itself gives  $\delta^i{}_i = \sum_i 1 = n = \dim V$ , so  $\text{tr } I = \dim V$ . The Kronecker delta is the identity operator, and contracted it is the dimension. ◀

The single operation that lowers tensor rank is *contraction*: summing over a matched upper and lower index,  $\tau^i{}_i$ , which for a  $(1, 1)$  tensor is exactly the trace. Under a change of basis each upper index transforms with  $P^{-1}$  and each lower index with  $P$  (Lecture 11), so a contracted pair is invariant.

**Example 12.22 (Contraction is the trace).** For an operator  $T = T^i{}_j e_i \otimes e^j$ , contracting the upper against the lower index gives  $T^i{}_i = \sum_i T^i{}_i = \text{tr } T$ , basis-independent by Lecture 11. More generally, contracting one pair of a higher tensor lowers its type by  $(1, 1)$ , turning, say, a  $(1, 2)$  tensor into a covector. ◀

**Example 12.23 (A transformation law).** A  $(1, 1)$  tensor transforms as  $T'^i{}_j = (P^{-1})^i{}_k T^k{}_l P^l{}_j$ , a  $(0, 2)$  tensor as  $g'_{ij} = P^k{}_i P^l{}_j g_{kl}$ , and a  $(2, 0)$  tensor with two factors of  $P^{-1}$ . The rule is mechanical: one factor of  $P^{-1}$  per upper index, one of  $P$  per lower index. ◀

**Example 12.24 (A  $(2, 0)$  tensor).** An antisymmetric  $(2, 0)$  tensor  $\omega^{ij} = -\omega^{ji}$  is a *bivector*; in  $\mathbb{R}^3$  its three independent components  $\omega^{12}, \omega^{23}, \omega^{31}$  are those of the cross product, which is why the cross product is really an antisymmetric tensor wearing the costume of a vector. ◀

## 4 Symmetric and antisymmetric tensors

Within  $V \otimes V$  live two natural subspaces.

**Definition 12.25 (Symmetric and antisymmetric tensors).** A tensor in  $V \otimes V$  is *symmetric* if swapping its two slots leaves it unchanged and *antisymmetric* if swapping flips the sign. These form subspaces  $S^2V$  and  $\Lambda^2V$ , and

$$V \otimes V = S^2V \oplus \Lambda^2V.$$

They are built from vectors by the symmetrized product  $v \otimes w + w \otimes v$  and the wedge  $v \wedge w = v \otimes w - w \otimes v$ .

The wedge product is alternating ( $v \wedge v = 0$ ) and feeds the theory of determinants and differential forms (Lecture 11). The split mirrors the decomposition of a square matrix into symmetric and antisymmetric parts. In physics the symmetric tensors describe bosons (symmetric multi-particle states) and the antisymmetric ones describe fermions (the Pauli exclusion principle is the antisymmetry  $v \wedge v = 0$ ).

**Example 12.26 (Splitting a tensor).** Any  $\tau = \sum \tau^{ij} e_i \otimes e_j$  splits as  $\tau = \tau_+ + \tau_-$  with symmetric part  $\tau_+^{ij} = \frac{1}{2}(\tau^{ij} + \tau^{ji})$  and antisymmetric part  $\tau_-^{ij} = \frac{1}{2}(\tau^{ij} - \tau^{ji})$ . For  $e_1 \otimes e_2$  this gives  $\frac{1}{2}(e_1 \otimes e_2 + e_2 \otimes e_1) + \frac{1}{2}e_1 \wedge e_2$ —a symmetric piece plus a wedge. ◀

**Example 12.27 (Symmetric tensors are polynomials).** Symmetric tensors in  $S^k V^*$  correspond to homogeneous degree- $k$  polynomials in the coordinates,  $e^i \otimes_S e^j \leftrightarrow x_i x_j$ . Their dimension  $\binom{n+k-1}{k}$  counts degree- $k$  monomials in  $n$  variables: symmetric tensor algebra is polynomial algebra. ◀

**Remark 12.28 (Identical particles).** Quantum statistics is the symmetry of multi-particle tensors:  $n$  identical bosons live in the symmetric power  $S^n \mathcal{H}$ ,  $n$  fermions in  $\Lambda^n \mathcal{H}$ . Since  $\Lambda^n$  vanishes once  $n$  exceeds  $\dim \mathcal{H}$ , no two fermions share a state—the Pauli principle, restated as a fact about antisymmetric tensors.

**Remark 12.29 (Dimensions).** For  $\dim V = n$ ,  $\dim S^2V = \binom{n+1}{2}$  and  $\dim \Lambda^2V = \binom{n}{2}$ , summing to  $n^2 = \dim(V \otimes V)$ . At the top,  $\Lambda^n V$  is one-dimensional—its single basis element is the determinant.

**Example 12.30 (The wedge and the determinant).** In  $\Lambda^2 \mathbb{R}^2$ ,  $u \wedge v = (u_1 v_2 - u_2 v_1) e_1 \wedge e_2$ : the lone coefficient is the  $2 \times 2$  determinant. In general

$$v_1 \wedge \cdots \wedge v_n = \det[v_1 \cdots v_n] (e_1 \wedge \cdots \wedge e_n),$$

so the determinant is the component of the top wedge—volume as an antisymmetric tensor. ◀

**Example 12.31 (Two-forms in three dimensions).**  $\Lambda^2 \mathbb{R}^3$  has dimension  $\binom{3}{2} = 3$ , with basis  $e_1 \wedge e_2, e_2 \wedge e_3, e_3 \wedge e_1$ —the same count as vectors in  $\mathbb{R}^3$ . This coincidence is why the cross

product exists only in three dimensions: there a bivector  $u \wedge v$  can be reinterpreted as a vector.  $\triangleleft$

## 5 Tensor products in quantum mechanics

The state space of a composite system is the tensor product of the parts:  $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ . For two qubits,  $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$  with orthonormal basis  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$  (writing  $|ij\rangle = |i\rangle \otimes |j\rangle$ ).

**Definition 12.32 (Product and entangled states).** A state of  $\mathcal{H}_A \otimes \mathcal{H}_B$  is a *product state* if it factors as  $|\psi\rangle \otimes |\phi\rangle$ , and *entangled* otherwise.

The Bell state  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  is entangled: no choice of single-qubit states  $|\psi\rangle, |\phi\rangle$  gives  $|\psi\rangle \otimes |\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . Operators on the composite are built as  $A \otimes B$  (act with  $A$  on the first factor,  $B$  on the second), though—crucially—not every operator is of this form, which is exactly how entangling gates generate entanglement from product inputs.

**Example 12.33 (The Bell state is entangled).** Suppose  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$ . Matching coefficients forces  $ad = bc = 0$  yet  $ac = bd = \frac{1}{\sqrt{2}} \neq 0$ , a contradiction. No product factorization exists—the state is entangled.  $\triangleleft$

**Example 12.34 (A state that does factor).** By contrast  $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle \otimes |+\rangle$ : its coefficient grid  $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  has rank one, so it is a product state, not entangled.  $\triangleleft$

**Example 12.35 (A decomposable tensor's grid).** The product  $(|0\rangle + |1\rangle) \otimes (|0\rangle + 2|1\rangle)$  expands to  $|00\rangle + 2|01\rangle + |10\rangle + 2|11\rangle$ , with coefficient grid

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix},$$

manifestly rank one—the algebraic signature of a product state.  $\triangleleft$

**Example 12.36 (Creating entanglement with a gate).** A Hadamard on the first qubit of  $|00\rangle$  gives the product state  $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ ; the controlled-NOT  $|1x\rangle \mapsto |1\bar{x}\rangle$  then yields  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . A product input has become the entangled Bell state.  $\triangleleft$

**Example 12.37 (The CNOT matrix).** In the basis  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$  the controlled-NOT is

$$U_{\text{CN}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

the identity on the first two basis states and a swap on the last two. It is *not* of the form  $A \otimes B$ , which is precisely why it can entangle.  $\triangleleft$

**Example 12.38 (Singlet and triplet).** Two spin- $\frac{1}{2}$  particles combine in  $\mathbb{C}^2 \otimes \mathbb{C}^2$  into the anti-symmetric *singlet*  $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$  and three symmetric *triplet* states. The singlet is maximally entangled, and the symmetry/antisymmetry under particle exchange is the  $S^2 \oplus \Lambda^2$  split of Definition 12.25 made physical.  $\triangleleft$

**Example 12.39 (Three-qubit entanglement).** The GHZ state  $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$  lives in  $(\mathbb{C}^2)^{\otimes 3} = \mathbb{C}^8$  and is genuinely three-party entangled: no splitting into a single qubit and a pair reproduces it. Composite systems pile up one tensor factor per subsystem.  $\triangleleft$

**Example 12.40 (A correlation).** On the Bell state, the joint observable  $\sigma_z \otimes \sigma_z$  has expectation +1—measuring  $\sigma_z$  on both qubits always gives equal outcomes—while the local expectations  $\sigma_z \otimes I$  and  $I \otimes \sigma_z$  are each 0. Perfect correlation with no local information: entanglement in a number. ◀

**Example 12.41 (Correlation in the  $x$ -basis).** On  $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , the flip-both operator  $\sigma_x \otimes \sigma_x$  of Example 12.7 gives

$$(\sigma_x \otimes \sigma_x)|\Phi\rangle = \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle) = |\Phi\rangle,$$

so  $\langle\Phi|\sigma_x \otimes \sigma_x|\Phi\rangle = 1$ : the Bell state is perfectly correlated in the  $x$ -basis as well as the  $z$ -basis. ◀

**Remark 12.42 (The Bell basis).** The four Bell states  $\frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$  and  $\frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$  form an orthonormal basis of  $\mathbb{C}^2 \otimes \mathbb{C}^2$  made entirely of maximally entangled states. A measurement in this basis—a *Bell measurement*—is the key step of teleportation and superdense coding.

**Example 12.43 (The four Bell states).** The Bell basis of  $\mathbb{C}^2 \otimes \mathbb{C}^2$  is

$$\begin{aligned} |\Phi^\pm\rangle &= \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \\ |\Psi^\pm\rangle &= \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle), \end{aligned}$$

four orthonormal, maximally entangled states. Every two-qubit state expands in this basis, and a Bell measurement projects onto one of them. ◀

**Remark 12.44 (Teleportation and no cloning).** A shared Bell pair lets one party send an unknown qubit using only a Bell measurement and two classical bits—entanglement plus the tensor-product structure moves quantum information without moving the particle (and not faster than light, since the classical bits are needed). Conversely, no linear  $U$  can satisfy  $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$  for all  $|\psi\rangle$ : unknown states cannot be cloned, a structural fact about operators on  $\mathcal{H} \otimes \mathcal{H}$ .

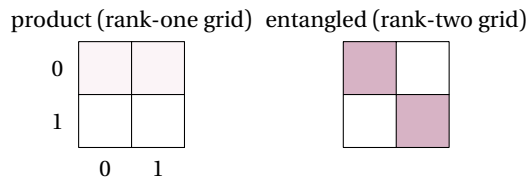


Figure 3: Coefficient grids  $\tau^{ij}$  of a two-qubit state. A product state's grid is rank one (an outer product of single-qubit amplitudes); the Bell state's grid is rank two and cannot be factored—the visual signature of entanglement.

**Remark 12.45 (Reduced states and the partial trace).** A subsystem of an entangled state has no pure state of its own; it is described by a *reduced density matrix*, the *partial trace* over the other factor (a contraction on one tensor slot). For a Bell state each qubit's reduced state is the maximally mixed  $\frac{1}{2}I$ —maximal ignorance of the part despite full knowledge of the whole, the hallmark of entanglement.

**Example 12.46 (A reduced density matrix).** For the Bell state  $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , tracing out the second qubit gives  $\rho_A = \text{tr}_B|\Phi\rangle\langle\Phi| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I$  on the first qubit. The part is maximally mixed even though the whole is a definite pure state—the defining tension of entanglement. ◀



**Remark 12.47 (Schmidt decomposition).** Every pure state of  $\mathcal{H}_A \otimes \mathcal{H}_B$  can be written  $\sum_k \lambda_k |a_k\rangle \otimes |b_k\rangle$  with orthonormal  $|a_k\rangle, |b_k\rangle$  and  $\lambda_k \geq 0$  (the *Schmidt decomposition*, the tensor-product cousin of the SVD). The count of nonzero  $\lambda_k$  is the Schmidt rank: 1 for product states, larger for entangled ones.

**Remark 12.48 (Exponential growth).** An  $n$ -qubit system has state space  $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$ : the dimension doubles with each qubit. This exponential growth is both the promise of quantum computing—a state holds  $2^n$  amplitudes—and the obstacle to simulating quantum systems on classical hardware.

### Summary of main concepts

The tensor product  $V \otimes W$  has basis  $\{e_i \otimes f_j\}$  and dimension  $(\dim V)(\dim W)$ ; its elements are sums of elementary tensors  $v \otimes w$ , most of which do not factor. In coordinates this is the Kronecker product, with  $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$ ,  $\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B$ , and eigenvalues  $\lambda\mu$ . A type- $(p, q)$  tensor is a multilinear map with  $p$  upper and  $q$  lower indices, unifying vectors, covectors, the metric, and operators ( $\mathcal{L}(V, W) \cong W \otimes V^*$ ); contraction over a matched pair generalizes the trace. Within  $V \otimes V$ , the symmetric and antisymmetric tensors split the space and model bosons and fermions. In quantum mechanics composite systems live in  $\mathcal{H}_A \otimes \mathcal{H}_B$ , and entanglement is the existence of states that do not factor.

- ▶ **Tensor product**  $V \otimes W$ —basis  $\{e_i \otimes f_j\}$ , dimension  $(\dim V)(\dim W)$ ; most elements do not factor.
- ▶ **Kronecker product**—the coordinate form;  $\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B$ , eigenvalues  $\lambda\mu$ , size  $nm \times nm$ .
- ▶ **Type- $(p, q)$  tensor**—a multilinear map with  $p$  upper and  $q$  lower indices; unifies vectors, covectors, the metric, and operators ( $\mathcal{L}(V, W) \cong W \otimes V^*$ ); contraction generalizes the trace.
- ▶ **Symmetric and antisymmetric tensors**—split  $V \otimes V$ ; model bosons and fermions.
- ▶ **Composite systems and entanglement**— $\mathcal{H}_A \otimes \mathcal{H}_B$ ; entangled states are those that do not factor.

*Exercises for this lecture are in the companion sheet `exercises-12.pdf`.*